

Details on the numerical experiments of

**SPARSE GRIDS VS. RANDOM POINTS
FOR HIGH-DIMENSIONAL
POLYNOMIAL APPROXIMATION**

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ABSTRACT. This document contains detailed information of the conducted numerical experiments in the paper [Eggl, Mindlberger, Ullrich: Sparse grids vs. random points for high-dimensional polynomial approximation].

		Scale 3		Scale 4		Scale 5		Scale 6		Scale 7		Scale 8		Scale 9	
		$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$
Bin. Gauss. $Q = 50$	Smolyak	5.71e-01	2.69e-01	3.66e-01	1.55e-01	1.91e-01	7.14e-02	3.33e-02	9.30e-03	2.47e-03	6.16e-04	8.96e-06	2.44e-06	7.95e-09	2.19e-09
	LS-Uniform	3.51e+00	8.16e-01	5.12e+00	7.82e-01	3.56e+00	3.27e-01	8.19e+00	7.13e-01	3.15e+02	1.41e+01	1.93e+00	6.95e-02	3.10e+04	5.42e+02
	LS-Chebyshev	1.74e+00	4.02e-01	4.13e-01	1.05e-01	1.73e-01	4.02e-02	3.74e-02	5.11e-03	1.55e-03	3.41e-04	5.78e-06	1.18e-06	7.91e-09	1.12e-09
Continuous $Q = 50$	Smolyak	4.11e-02	1.25e-02	2.36e-02	5.02e-03	1.01e-02	1.52e-03	5.52e-03	5.30e-04	2.86e-03	2.38e-04	1.39e-03	6.77e-05	6.34e-04	2.43e-05
	LS-Uniform	3.74e-01	9.56e-02	6.27e-01	9.21e-02	2.60e-01	2.32e-02	4.02e-01	3.62e-02	1.02e+03	4.27e+01	7.11e+02	2.94e+01	1.43e+09	2.51e+07
	LS-Chebyshev	1.30e-01	3.14e-02	1.55e-02	4.18e-03	9.59e-03	1.69e-03	5.82e-03	7.29e-04	2.30e-03	2.32e-04	1.19e-03	7.56e-05	6.40e-04	2.54e-05
Corner Peak $Q = 50$	Smolyak	8.78e-03	4.52e-03	1.48e-03	5.81e-04	1.04e-04	3.31e-05	8.90e-07	2.61e-07	6.08e-09	1.75e-09	2.77e-13	8.54e-14	6.99e-15	9.35e-16
	LS-Uniform	1.54e-02	3.24e-03	3.50e-03	5.50e-04	4.92e-04	5.98e-05	2.30e-04	2.04e-05	6.80e-04	3.05e-05	8.10e-08	2.71e-09	2.76e-02	4.87e-04
	LS-Chebyshev	2.14e-02	4.71e-03	1.02e-03	2.11e-04	5.74e-05	1.15e-05	8.85e-07	1.18e-07	4.29e-09	8.04e-10	3.04e-13	3.51e-14	7.75e-15	3.77e-16
Discont. $Q = 50$	Smolyak	4.73e+00	1.26e+00	3.82e+00	1.08e+00	4.82e+00	8.42e-01	4.56e+00	6.75e-01	4.07e+00	4.26e-01	3.57e+00	3.28e-01	4.70e+00	2.16e-01
	LS-Uniform	3.46e+01	8.99e+00	7.48e+01	1.09e+01	6.02e+01	6.33e+00	3.07e+02	2.72e+01	6.62e+05	2.77e+04	1.79e+06	6.20e+04	6.74e+12	1.18e+11
	LS-Chebyshev	7.38e+00	1.53e+00	3.73e+00	8.73e-01	3.36e+00	6.26e-01	3.51e+00	4.21e-01	3.10e+00	3.16e-01	2.86e+00	2.37e-01	3.25e+00	1.77e-01
Gaussian $Q = 50$	Smolyak	5.81e-01	2.39e-01	3.51e-01	9.72e-02	1.18e-01	3.11e-02	2.17e-02	5.03e-03	1.93e-03	4.27e-04	7.40e-06	1.88e-06	6.31e-09	1.51e-09
	LS-Uniform	3.13e+00	7.97e-01	3.83e+00	5.77e-01	3.44e+00	2.92e-01	7.53e+00	6.71e-01	2.27e+02	1.11e+01	2.62e+00	8.97e-02	1.97e+04	3.46e+02
	LS-Chebyshev	2.05e+00	5.21e-01	3.30e-01	6.64e-02	9.31e-02	1.92e-02	1.91e-02	2.94e-03	1.13e-03	2.37e-04	5.22e-06	9.35e-07	6.92e-09	7.79e-10
Geo. Mean $Q = 50$	Smolyak	1.09e-02	5.50e-03	2.80e-03	7.19e-04	3.70e-04	7.25e-05	4.84e-05	8.72e-06	3.65e-06	6.94e-07	1.36e-07	2.22e-08	2.19e-09	3.61e-10
	LS-Uniform	7.18e-03	1.52e-03	5.70e-03	1.08e-03	1.33e-03	1.36e-04	3.29e-03	2.95e-04	1.07e-01	4.54e-03	3.88e-03	1.23e-04	2.27e+02	3.97e+00
	LS-Chebyshev	1.26e-02	3.54e-03	5.71e-03	7.35e-04	1.90e-04	3.79e-05	1.70e-05	2.97e-06	1.40e-06	2.32e-07	9.30e-08	7.28e-09	1.67e-09	1.17e-10
Noise $Q = 50$	Smolyak	5.40e-07	2.66e-07	6.29e-07	2.54e-07	7.82e-07	2.42e-07	9.42e-07	2.39e-07	9.59e-07	2.39e-07	1.07e-06	2.49e-07	1.22e-06	2.57e-07
	LS-Uniform	3.41e-06	8.64e-07	1.41e-05	2.10e-06	2.39e-05	2.22e-06	2.15e-04	1.90e-05	5.50e-01	3.03e-02	1.59e+00	4.81e-02	9.24e+06	1.62e+05
	LS-Chebyshev	2.46e-06	5.81e-07	7.67e-07	1.71e-07	6.78e-07	1.43e-07	6.45e-07	1.32e-07	5.78e-07	1.18e-07	6.13e-07	1.15e-07	9.04e-07	1.12e-07
Oscillatory $Q = 50$	Smolyak	3.97e-02	2.52e-02	2.12e-03	1.06e-03	4.48e-05	1.87e-05	9.42e-09	3.35e-09	5.70e-13	1.95e-13	9.44e-15	2.26e-15	1.97e-14	3.65e-15
	LS-Uniform	1.69e-01	3.71e-02	9.31e-03	1.42e-03	1.97e-04	3.00e-05	2.71e-06	2.40e-07	6.89e-08	3.52e-09	8.81e-09	2.73e-10	8.14e-02	1.43e-03
	LS-Chebyshev	9.09e-02	2.02e-02	1.92e-03	4.56e-04	3.64e-05	7.71e-06	1.16e-08	1.80e-09	4.78e-13	9.33e-14	9.33e-15	1.21e-15	2.74e-14	1.17e-15
Prod. Peak $Q = 50$	Smolyak	1.04e-03	6.17e-04	8.63e-05	3.77e-05	3.51e-06	1.24e-06	2.27e-08	7.41e-09	3.03e-11	8.55e-12	7.77e-15	2.15e-15	1.80e-14	3.63e-15
	LS-Uniform	4.27e-03	9.86e-04	4.65e-04	7.06e-05	1.99e-05	2.55e-06	7.28e-06	6.43e-07	3.65e-06	1.54e-07	3.99e-09	1.29e-10	3.00e-02	5.25e-04
	LS-Chebyshev	2.59e-03	5.70e-04	7.59e-05	1.69e-05	3.21e-06	5.62e-07	2.25e-08	3.53e-09	2.35e-11	4.32e-12	6.94e-15	7.72e-16	4.33e-15	7.07e-16
Ridge Prod. $Q = 50$	Smolyak	2.48e-01	1.05e-01	1.14e-01	3.85e-02	6.89e-02	1.54e-02	3.42e-02	5.39e-03	2.39e-02	2.13e-03	1.18e-02	9.11e-04	7.14e-03	3.33e-04
	LS-Uniform	1.29e+00	3.18e-01	2.37e+00	3.45e-01	1.61e+00	1.67e-01	9.01e+00	7.85e-01	6.39e+03	2.86e+02	5.10e+03	1.54e+02	1.57e+10	2.74e+08
	LS-Chebyshev	1.19e+00	2.57e-01	1.54e-01	3.16e-02	7.06e-02	1.38e-02	2.87e-02	4.10e-03	1.98e-02	1.70e-03	7.97e-03	6.06e-04	4.39e-03	2.27e-04

TABLE 1. The errors $e_{\max}^{\text{wc}}$ and $e_{\text{mean}}^{\text{wc}}$ for $d = 2$, each scale for the different algorithms. Lowest value in bold.

		Scale 3		Scale 4		Scale 5		Scale 6		Scale 7		Scale 8		Scale 9	
		$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$
Bin. Gauss. $Q = 50$	Smolyak	7.17e-01	2.93e-01	4.39e-01	1.60e-01	2.76e-01	7.71e-02	9.61e-02	2.28e-02	3.75e-02	6.63e-03	9.20e-03	1.35e-03	8.27e-04	1.15e-04
	LS-Uniform	2.50e+00	3.95e-01	7.74e-01	1.53e-01	1.26e+00	1.25e-01	7.64e-01	3.99e-02	9.90e-01	2.53e-02	2.32e+00	3.70e-02	4.82e+01	4.68e-01
	LS-Chebyshev	8.53e-01	2.44e-01	5.41e-01	1.36e-01	2.73e-01	5.43e-02	4.18e-02	1.14e-02	1.25e-02	2.04e-03	3.64e-03	3.97e-04	6.37e-04	3.67e-05
Continuous $Q = 50$	Smolyak	4.41e-02	1.11e-02	2.05e-02	4.18e-03	1.18e-02	1.37e-03	5.89e-03	5.26e-04	2.93e-03	1.84e-04	1.39e-03	6.69e-05	6.85e-04	2.36e-05
	LS-Uniform	9.44e-02	1.75e-02	2.98e-02	5.42e-03	4.18e-02	3.20e-03	4.80e-02	1.85e-03	9.59e-02	2.50e-03	2.29e-01	3.66e-03	2.96e+01	2.88e-01
	LS-Chebyshev	3.41e-02	9.63e-03	1.95e-02	5.55e-03	1.04e-02	1.72e-03	4.58e-03	5.82e-04	2.62e-03	1.93e-04	1.03e-03	6.84e-05	5.39e-04	2.45e-05
Corner Peak $Q = 50$	Smolyak	5.78e-02	1.58e-02	3.03e-02	5.80e-03	7.46e-03	1.46e-03	1.44e-03	2.55e-04	1.98e-04	2.96e-05	1.35e-05	2.06e-06	3.16e-07	4.17e-08
	LS-Uniform	5.83e-02	1.20e-02	8.83e-03	1.65e-03	8.35e-03	4.34e-04	2.03e-03	9.31e-05	1.40e-03	3.22e-05	6.45e-04	1.13e-05	5.31e-03	5.72e-05
	LS-Chebyshev	3.02e-02	5.61e-03	7.48e-03	1.31e-03	1.13e-03	2.09e-04	2.74e-04	2.78e-05	3.55e-05	3.82e-06	5.60e-06	4.14e-07	1.34e-07	8.16e-09
Discont. $Q = 50$	Smolyak	9.82e+00	1.73e+00	1.11e+01	1.44e+00	7.73e+00	1.03e+00	9.26e+00	8.10e-01	9.98e+00	5.76e-01	8.40e+00	4.81e-01	1.07e+01	3.70e-01
	LS-Uniform	1.70e+01	3.87e+00	1.14e+01	1.66e+00	2.38e+01	1.71e+00	6.09e+01	2.57e+00	2.28e+02	5.88e+00	1.77e+03	2.70e+01	4.47e+05	4.45e+03
	LS-Chebyshev	7.39e+00	2.01e+00	7.39e+00	1.44e+00	8.08e+00	1.02e+00	7.86e+00	7.12e-01	9.51e+00	5.15e-01	6.64e+00	3.78e-01	6.30e+00	2.68e-01
Gaussian $Q = 50$	Smolyak	6.32e-01	2.31e-01	4.62e-01	1.21e-01	2.07e-01	4.28e-02	9.50e-02	1.72e-02	3.17e-02	5.10e-03	6.88e-03	8.38e-04	7.69e-04	1.08e-04
	LS-Uniform	2.65e+00	4.06e-01	9.28e-01	9.70e-02	7.69e-01	6.15e-02	8.92e-01	3.17e-02	6.14e-01	1.82e-02	1.82e+00	2.80e-02	6.74e+01	6.18e-01
	LS-Chebyshev	4.76e-01	1.80e-01	3.31e-01	7.38e-02	1.67e-01	2.84e-02	4.48e-02	6.44e-03	1.14e-02	1.45e-03	2.53e-03	2.55e-04	3.99e-04	3.04e-05
Gm. Mean $Q = 50$	Smolyak	4.13e-02	1.15e-02	1.16e-02	2.16e-03	2.22e-03	3.94e-04	4.17e-04	5.71e-05	4.98e-05	5.56e-06	4.29e-06	3.94e-07	1.94e-07	1.71e-08
	LS-Uniform	4.56e-02	6.09e-03	9.33e-03	1.19e-03	1.24e-03	1.53e-04	4.69e-04	2.79e-05	5.67e-04	1.29e-05	3.38e-05	1.11e-06	8.01e-04	8.20e-06
	LS-Chebyshev	3.53e-02	7.18e-03	3.88e-03	8.24e-04	6.25e-04	1.13e-04	1.70e-04	1.58e-05	2.39e-05	1.69e-06	2.24e-06	1.14e-07	1.03e-07	4.99e-09
Noise $Q = 50$	Smolyak	8.06e-07	3.02e-07	1.18e-06	3.47e-07	1.42e-06	3.81e-07	1.83e-06	4.21e-07	2.43e-06	4.64e-07	2.90e-06	5.04e-07	3.00e-06	5.42e-07
	LS-Uniform	1.98e-06	3.83e-07	1.45e-06	2.08e-07	2.65e-06	2.86e-07	2.31e-05	8.05e-07	7.11e-05	2.09e-06	9.56e-04	1.49e-05	2.56e-01	2.72e-03
	LS-Chebyshev	5.14e-07	1.67e-07	1.02e-06	1.59e-07	1.04e-06	1.42e-07	9.48e-07	1.29e-07	1.01e-06	1.22e-07	9.65e-07	1.17e-07	1.35e-06	1.13e-07
Oscillatory $Q = 50$	Smolyak	4.60e-01	1.54e-01	8.45e-02	3.12e-02	9.48e-03	3.25e-03	5.95e-04	1.78e-04	2.00e-05	5.05e-06	2.81e-07	6.51e-08	7.89e-11	1.71e-11
	LS-Uniform	1.11e+00	2.39e-01	1.20e-01	1.66e-02	2.05e-02	1.94e-03	2.17e-03	1.11e-04	2.75e-04	7.39e-06	4.46e-05	7.80e-07	4.07e-06	3.67e-08
	LS-Chebyshev	3.74e-01	1.08e-01	1.34e-01	1.42e-02	5.92e-03	8.55e-04	3.11e-04	3.08e-05	8.41e-06	9.18e-07	1.33e-07	1.49e-08	6.44e-11	3.93e-12
Prod. Peak $Q = 50$	Smolyak	2.63e-03	1.01e-03	3.52e-04	1.21e-04	3.76e-05	1.21e-05	3.67e-06	9.19e-07	2.29e-07	4.46e-08	6.61e-09	1.15e-09	4.20e-11	7.27e-12
	LS-Uniform	7.50e-03	1.23e-03	4.61e-04	6.99e-05	8.53e-05	7.61e-06	7.87e-06	6.14e-07	2.64e-06	6.73e-08	8.02e-07	1.45e-08	1.46e-06	1.47e-08
	LS-Chebyshev	1.95e-03	6.82e-04	4.64e-04	5.65e-05	2.47e-05	3.75e-06	1.60e-06	1.83e-07	8.79e-08	8.99e-09	2.25e-09	2.70e-10	2.89e-11	1.75e-12
Ridge Prod. $Q = 50$	Smolyak	6.03e-01	1.98e-01	4.10e-01	8.99e-02	1.91e-01	4.03e-02	1.10e-01	1.54e-02	5.22e-02	6.44e-03	3.81e-02	2.47e-03	2.34e-02	9.64e-04
	LS-Uniform	1.08e+00	2.25e-01	3.92e-01	6.36e-02	4.15e-01	4.25e-02	9.80e-01	3.95e-02	1.56e+00	4.47e-02	8.14e+00	1.27e-01	7.94e-02	9.33e+00
	LS-Chebyshev	4.59e-01	1.16e-01	3.48e-01	5.84e-02	1.51e-01	2.13e-02	6.15e-02	9.06e-03	4.02e-02	3.63e-03	1.93e-02	1.35e-03	1.00e-02	5.45e-04

TABLE 2. The errors $e_{\max}^{\text{wc}}$ and $e_{\text{mean}}^{\text{wc}}$ for $d = 3$, each scale for the different algorithms. Lowest value in bold.

		Scale 3		Scale 4		Scale 5		Scale 6		Scale 7		Scale 8		Scale 9	
		$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$
Bin. Gauss. $Q = 50$	Smolyak	8.72e-01	2.38e-01	5.32e-01	1.24e-01	3.54e-01	6.76e-02	2.15e-01	3.79e-02	9.04e-02	1.18e-02	3.27e-02	4.77e-03	1.22e-02	1.35e-03
	LS-Uniform	1.47e+00	1.83e-01	1.43e+00	1.24e-01	1.11e+00	8.09e-02	1.37e+00	3.85e-02	5.60e-01	1.40e-02	4.67e-01	7.44e-03	1.62e+00	1.19e-02
	LS-Chebyshev	9.84e-01	2.00e-01	4.36e-01	8.53e-02	2.03e-01	4.78e-02	1.14e-01	1.84e-02	4.00e-02	3.88e-03	2.48e-02	1.10e-03	7.91e-03	2.31e-04
Continuous $Q = 50$	Smolyak	3.06e-02	8.31e-03	1.82e-02	4.05e-03	8.09e-03	1.23e-03	6.36e-03	4.41e-04	2.55e-03	1.83e-04	1.39e-03	5.56e-05	6.26e-04	2.00e-05
	LS-Uniform	4.78e-02	9.23e-03	3.60e-02	3.99e-03	2.92e-02	1.84e-03	1.68e-02	7.48e-04	1.35e-02	3.50e-04	2.49e-02	3.02e-04	1.41e-01	1.23e-03
	LS-Chebyshev	2.66e-02	8.04e-03	1.72e-02	3.13e-03	8.35e-03	1.13e-03	5.17e-03	4.27e-04	2.31e-03	1.62e-04	1.15e-03	5.83e-05	4.60e-04	2.07e-05
Corner Peak $Q = 50$	Smolyak	2.14e-01	3.12e-02	8.52e-02	1.57e-02	5.55e-02	5.66e-03	1.91e-02	2.31e-03	6.40e-03	6.41e-04	1.63e-03	1.36e-04	3.09e-04	2.19e-05
	LS-Uniform	4.68e-02	8.95e-03	3.39e-02	2.42e-03	3.20e-02	1.49e-03	8.57e-03	2.94e-04	7.99e-03	1.05e-04	1.63e-03	2.31e-05	3.87e-03	2.77e-05
	LS-Chebyshev	7.48e-02	8.17e-03	2.88e-02	2.24e-03	6.40e-03	6.17e-04	3.11e-03	1.42e-04	1.07e-03	3.04e-05	1.63e-04	5.30e-06	3.82e-05	1.01e-06
Discont. $Q = 50$	Smolyak	3.89e+01	4.96e+00	3.28e+01	4.64e+00	4.90e+01	4.26e+00	4.48e+01	2.64e+00	3.96e+01	1.97e+00	3.72e+01	1.50e+00	3.72e+01	1.23e+00
	LS-Uniform	3.74e+01	5.14e+00	3.30e+01	4.56e+00	6.67e+01	5.43e+00	8.11e+01	3.80e+00	1.09e+02	3.34e+00	5.45e+02	6.40e+00	6.77e+03	4.80e+01
	LS-Chebyshev	2.38e+01	5.40e+00	2.94e+01	3.72e+00	2.85e+01	3.00e+00	2.71e+01	2.16e+00	2.19e+01	1.58e+00	2.12e+01	1.10e+00	2.15e+01	7.76e-01
Gaussian $Q = 50$	Smolyak	6.68e-01	1.36e-01	5.95e-01	7.86e-02	3.89e-01	4.75e-02	2.00e-01	2.17e-02	7.77e-02	8.03e-03	3.33e-02	2.83e-03	1.09e-02	8.58e-04
	LS-Uniform	8.17e-01	1.22e-01	8.59e-01	6.87e-02	8.47e-01	4.22e-02	7.31e-01	1.91e-02	3.24e-01	8.47e-03	2.36e-01	4.40e-03	1.30e+00	8.91e-03
	LS-Chebyshev	5.07e-01	9.84e-02	3.42e-01	5.65e-02	1.47e-01	2.35e-02	6.97e-02	8.07e-03	2.95e-02	2.81e-03	1.90e-02	7.96e-04	4.08e-03	1.57e-04
Geo. Mean $Q = 50$	Smolyak	3.86e-02	8.69e-03	8.81e-03	1.39e-03	1.59e-03	2.04e-04	1.85e-04	2.45e-05	2.15e-05	2.16e-06	1.75e-06	1.27e-07	7.27e-08	4.95e-09
	LS-Uniform	5.11e-02	6.36e-03	1.10e-02	8.16e-04	1.46e-03	1.01e-04	2.33e-04	1.04e-05	1.42e-04	2.22e-06	1.29e-05	1.80e-07	2.56e-06	2.13e-08
	LS-Chebyshev	3.71e-02	6.80e-03	4.12e-03	6.58e-04	1.23e-03	7.67e-05	6.83e-05	7.27e-06	2.63e-05	7.03e-07	1.00e-06	3.86e-08	2.76e-08	1.22e-09
Noise $Q = 50$	Smolyak	1.29e-06	4.28e-07	2.06e-06	5.22e-07	2.96e-06	5.49e-07	3.53e-06	6.58e-07	4.47e-06	7.63e-07	5.78e-06	8.89e-07	7.84e-06	1.03e-06
	LS-Uniform	1.08e-06	1.83e-07	2.25e-06	2.07e-07	5.62e-06	2.49e-07	1.44e-05	3.97e-07	1.36e-05	4.53e-07	9.45e-05	1.06e-06	1.79e-03	1.11e-05
	LS-Chebyshev	6.05e-07	1.67e-07	7.11e-07	1.45e-07	7.48e-07	1.30e-07	9.21e-07	1.25e-07	1.31e-06	1.23e-07	8.93e-07	1.20e-07	1.14e-06	1.18e-07
Oscillatory $Q = 50$	Smolyak	1.89e+00	3.73e-01	6.08e-01	1.40e-01	1.44e-01	3.66e-02	2.14e-02	6.13e-03	2.37e-03	6.07e-04	1.67e-04	3.55e-05	7.13e-06	1.26e-06
	LS-Uniform	2.76e+00	4.92e-01	1.21e+00	1.10e-01	4.07e-01	2.28e-02	9.78e-02	2.34e-03	4.72e-03	1.57e-04	1.11e-03	1.61e-05	5.99e-04	4.26e-06
	LS-Chebyshev	1.33e+00	3.58e-01	8.72e-01	9.22e-02	1.29e-01	1.37e-02	1.32e-02	1.17e-03	1.01e-03	6.79e-05	7.70e-05	3.04e-06	3.47e-06	1.01e-07
Prod. Peak $Q = 50$	Smolyak	5.72e-03	1.21e-03	7.78e-04	2.11e-04	1.33e-04	3.31e-05	1.82e-05	4.20e-06	1.90e-06	4.52e-07	1.69e-07	3.66e-08	1.26e-08	2.03e-09
	LS-Uniform	5.29e-03	1.03e-03	1.56e-03	1.40e-04	2.71e-04	2.11e-05	7.25e-05	1.72e-06	4.65e-06	1.73e-07	1.22e-06	2.24e-08	6.35e-07	6.32e-09
	LS-Chebyshev	4.61e-03	1.06e-03	9.07e-04	1.06e-04	7.52e-05	1.05e-05	1.18e-05	9.48e-07	5.73e-07	6.83e-08	7.15e-08	4.08e-09	5.58e-09	1.98e-10
Ridge Prod. $Q = 50$	Smolyak	1.42e+00	3.50e-01	8.23e-01	1.75e-01	5.84e-01	8.02e-02	2.91e-01	3.56e-02	1.79e-01	1.52e-02	9.41e-02	6.41e-03	4.74e-02	2.67e-03
	LS-Uniform	2.28e+00	2.94e-01	1.12e+00	1.14e-01	1.21e+00	7.49e-02	1.89e+00	4.70e-02	4.97e-01	1.95e-02	2.27e+00	2.50e-02	1.41e+01	1.02e-01
	LS-Chebyshev	1.02e+00	2.15e-01	5.51e-01	9.66e-02	2.52e-01	4.35e-02	1.72e-01	1.70e-02	9.43e-02	7.00e-03	5.68e-02	2.98e-03	2.71e-02	1.20e-03

TABLE 3. The errors $e_{\max}^{\text{wc}}$ and $e_{\text{mean}}^{\text{wc}}$ for $d = 4$, each scale for the different algorithms. Lowest value in bold.

		Scale 3		Scale 4		Scale 5		Scale 6		Scale 7		Scale 8	
		$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$
Bin. Gauss. $Q = 50$	Smolyak	6.68e-01	1.75e-01	5.97e-01	9.89e-02	4.13e-01	6.23e-02	2.65e-01	2.86e-02	1.17e-01	1.47e-02	5.25e-02	5.69e-03
	LS-Uniform	7.19e-01	1.46e-01	1.16e+00	1.10e-01	1.10e+00	5.32e-02	7.42e-01	2.94e-02	1.05e+00	1.21e-02	6.47e-01	6.31e-03
	LS-Chebyshev	6.94e-01	1.66e-01	4.42e-01	8.76e-02	2.64e-01	4.13e-02	1.16e-01	1.73e-02	5.47e-02	5.48e-03	2.51e-02	1.38e-03
Continuous $Q = 50$	Smolyak	2.96e-02	8.36e-03	1.54e-02	2.71e-03	6.89e-03	9.20e-04	4.43e-03	3.35e-04	1.66e-03	1.23e-04	1.08e-03	4.65e-05
	LS-Uniform	3.81e-02	7.90e-03	1.90e-02	3.25e-03	1.30e-02	1.36e-03	1.29e-02	5.90e-04	8.18e-03	2.47e-04	7.69e-03	1.14e-04
	LS-Chebyshev	2.92e-02	7.51e-03	1.41e-02	3.17e-03	7.32e-03	1.05e-03	3.52e-03	3.67e-04	1.72e-03	1.35e-04	8.18e-04	4.84e-05
Corner Peak $Q = 50$	Smolyak	2.02e-01	3.27e-02	1.71e-01	2.45e-02	1.16e-01	1.05e-02	7.70e-02	5.35e-03	4.09e-02	2.23e-03	1.68e-02	9.96e-04
	LS-Uniform	4.61e-02	4.52e-03	8.67e-02	3.27e-03	5.12e-02	1.32e-03	1.69e-02	4.12e-04	1.82e-02	1.85e-04	9.44e-03	5.31e-05
	LS-Chebyshev	5.92e-02	5.49e-03	4.61e-02	2.48e-03	1.31e-02	9.41e-04	1.19e-02	2.97e-04	3.93e-03	7.39e-05	1.83e-03	1.90e-05
Discont. $Q = 50$	Smolyak	5.38e+01	7.83e+00	5.14e+01	5.47e+00	4.26e+01	4.36e+00	4.60e+01	3.59e+00	5.27e+01	2.32e+00	5.04e+01	1.95e+00
	LS-Uniform	6.21e+01	9.16e+00	5.76e+01	7.68e+00	5.02e+01	5.99e+00	1.05e+02	4.97e+00	1.53e+02	4.95e+00	3.00e+02	4.55e+00
	LS-Chebyshev	4.37e+01	8.34e+00	4.01e+01	5.62e+00	4.68e+01	4.43e+00	4.61e+01	3.06e+00	4.31e+01	2.37e+00	5.28e+01	1.77e+00
Gaussian $Q = 50$	Smolyak	6.99e-01	1.59e-01	6.26e-01	8.24e-02	3.62e-01	4.76e-02	2.10e-01	1.99e-02	9.82e-02	8.04e-03	4.12e-02	4.09e-03
	LS-Uniform	6.77e-01	8.96e-02	8.02e-01	6.98e-02	1.06e+00	3.57e-02	7.78e-01	1.58e-02	7.32e-01	9.70e-03	8.41e-01	4.98e-03
	LS-Chebyshev	5.02e-01	8.80e-02	3.21e-01	4.27e-02	2.62e-01	1.95e-02	1.17e-01	9.25e-03	4.39e-02	3.36e-03	1.69e-02	1.10e-03
Geo. Mean $Q = 50$	Smolyak	1.22e-01	3.26e-02	5.09e-02	7.90e-03	2.18e-02	1.82e-03	6.24e-03	3.31e-04	1.13e-03	5.22e-05	2.10e-04	7.33e-06
	LS-Uniform	1.21e-01	1.44e-02	3.30e-02	3.10e-03	1.05e-02	8.07e-04	1.33e-02	2.31e-04	1.92e-03	2.70e-05	5.42e-04	5.35e-06
	LS-Chebyshev	8.56e-02	1.80e-02	2.22e-02	3.97e-03	7.27e-03	6.82e-04	4.18e-03	1.20e-04	5.48e-04	1.37e-05	4.26e-05	1.60e-06
Noise $Q = 50$	Smolyak	2.40e-06	5.09e-07	2.99e-06	6.76e-07	4.15e-06	8.23e-07	5.77e-06	9.69e-07	9.79e-06	1.22e-06	1.08e-05	1.45e-06
	LS-Uniform	1.29e-06	1.65e-07	1.76e-06	1.82e-07	3.06e-06	1.97e-07	7.48e-06	2.36e-07	1.43e-05	2.99e-07	2.91e-05	4.29e-07
	LS-Chebyshev	7.36e-07	1.57e-07	7.38e-07	1.43e-07	9.82e-07	1.37e-07	1.09e-06	1.30e-07	1.11e-06	1.29e-07	1.45e-06	1.24e-07
Oscillatory $Q = 50$	Smolyak	2.95e+00	6.93e-01	2.05e+00	3.44e-01	7.39e-01	1.35e-01	2.06e-01	3.66e-02	4.03e-02	8.20e-03	5.78e-03	1.13e-03
	LS-Uniform	8.65e+00	9.19e-01	4.22e+00	3.33e-01	2.14e+00	9.46e-02	1.28e+00	2.24e-02	2.04e-01	3.08e-03	3.25e-02	3.72e-04
	LS-Chebyshev	3.88e+00	8.35e-01	1.67e+00	2.77e-01	7.48e-01	7.01e-02	2.10e-01	1.10e-02	2.76e-02	1.36e-03	3.26e-03	1.09e-04
Prod. Peak $Q = 50$	Smolyak	1.05e-02	1.27e-03	1.03e-03	2.41e-04	2.12e-04	3.60e-05	2.62e-05	4.65e-06	3.23e-06	5.04e-07	3.41e-07	4.08e-08
	LS-Uniform	8.47e-03	1.16e-03	2.30e-03	2.04e-04	7.37e-04	2.35e-05	6.91e-05	2.06e-06	1.29e-05	2.30e-07	4.36e-06	2.48e-08
	LS-Chebyshev	6.41e-03	1.09e-03	1.42e-03	1.44e-04	1.30e-04	1.50e-05	1.50e-05	1.30e-06	1.62e-06	1.01e-07	2.85e-07	7.96e-09
Ridge Prod. $Q = 50$	Smolyak	3.49e+00	6.50e-01	2.37e+00	3.23e-01	1.63e+00	1.67e-01	8.97e-01	7.89e-02	4.30e-01	3.54e-02	2.17e-01	1.58e-02
	LS-Uniform	2.43e+00	4.34e-01	2.98e+00	2.05e-01	1.94e+00	1.01e-01	1.67e+00	5.02e-02	1.04e+00	3.15e-02	1.39e+00	1.87e-02
	LS-Chebyshev	1.58e+00	3.54e-01	1.21e+00	1.76e-01	6.83e-01	7.75e-02	3.13e-01	3.08e-02	2.01e-01	1.34e-02	1.07e-01	5.67e-03

TABLE 4. The errors $e_{\max}^{\text{wc}}$ and $e_{\text{mean}}^{\text{wc}}$ for $d = 5$, each scale for the different algorithms. Lowest value in bold.

			Scale 3		Scale 4		Scale 5		Scale 6		Scale 7	
			$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$
Bim. Gauss. $Q = 50$	Smolyak		6.95e-01	1.90e-01	5.82e-01	1.05e-01	4.37e-01	5.67e-02	2.66e-01	3.20e-02	1.55e-01	1.64e-02
	LS-Uniform		6.99e-01	1.01e-01	6.72e-01	7.61e-02	8.14e-01	3.69e-02	2.04e+00	2.48e-02	9.83e-01	1.14e-02
	LS-Chebyshev		6.07e-01	1.13e-01	4.77e-01	6.17e-02	2.38e-01	2.87e-02	2.57e-01	1.23e-02	8.82e-02	4.88e-03
Continuous $Q = 50$	Smolyak		2.47e-02	5.38e-03	1.47e-02	2.24e-03	7.88e-03	7.95e-04	3.64e-03	3.16e-04	1.66e-03	1.05e-04
	LS-Uniform		3.45e-02	6.70e-03	2.30e-02	2.28e-03	1.15e-02	9.45e-04	1.40e-02	4.08e-04	8.73e-03	1.68e-04
	LS-Chebyshev		2.44e-02	7.29e-03	1.51e-02	2.19e-03	5.85e-03	8.47e-04	2.31e-03	3.03e-04	1.18e-03	1.11e-04
Corner Peak $Q = 50$	Smolyak		1.36e-01	1.60e-02	1.76e-01	1.83e-02	1.63e-01	1.01e-02	1.06e-01	5.48e-03	1.06e-01	2.83e-03
	LS-Uniform		2.17e-02	4.52e-03	1.21e-01	3.21e-03	9.20e-02	1.34e-03	3.10e-02	4.09e-04	3.23e-02	2.13e-04
	LS-Chebyshev		1.80e-02	4.15e-03	6.08e-02	2.26e-03	3.18e-02	1.03e-03	1.21e-02	3.69e-04	8.72e-03	1.31e-04
Discont. $Q = 50$	Smolyak		1.46e+02	1.59e+01	1.18e+02	1.46e+01	1.69e+02	1.17e+01	1.77e+02	8.48e+00	1.62e+02	6.02e+00
	LS-Uniform		1.33e+02	1.97e+01	1.26e+02	1.38e+01	1.55e+02	1.01e+01	2.41e+02	9.99e+00	3.79e+02	8.41e+00
	LS-Chebyshev		1.13e+02	1.89e+01	1.43e+02	1.35e+01	1.12e+02	8.77e+00	1.16e+02	6.51e+00	1.12e+02	5.02e+00
Gaussian $Q = 50$	Smolyak		6.21e-01	1.01e-01	6.68e-01	7.20e-02	4.76e-01	4.67e-02	3.58e-01	2.35e-02	1.91e-01	1.42e-02
	LS-Uniform		9.36e-01	8.52e-02	5.42e-01	4.77e-02	7.09e-01	2.81e-02	1.83e+00	2.17e-02	1.18e+00	8.96e-03
	LS-Chebyshev		5.55e-01	9.55e-02	3.68e-01	4.07e-02	2.54e-01	2.27e-02	3.09e-01	1.11e-02	1.10e-01	4.40e-03
Geo. Mean $Q = 50$	Smolyak		7.75e-02	1.67e-02	3.03e-02	3.31e-03	6.80e-03	5.58e-04	1.36e-03	8.65e-05	2.06e-04	1.15e-05
	LS-Uniform		8.41e-02	1.05e-02	2.16e-02	1.77e-03	5.45e-03	2.24e-04	1.61e-03	3.74e-05	2.72e-04	4.62e-06
	LS-Chebyshev		5.53e-02	1.12e-02	1.30e-02	1.75e-03	2.81e-03	2.22e-04	5.74e-04	2.46e-05	6.13e-05	2.90e-06
Noise $Q = 50$	Smolyak		3.47e-06	6.21e-07	5.26e-06	8.69e-07	7.74e-06	1.17e-06	1.10e-05	1.47e-06	1.43e-05	1.80e-06
	LS-Uniform		1.14e-06	1.64e-07	1.76e-06	1.59e-07	3.03e-06	1.70e-07	8.56e-06	2.13e-07	1.34e-05	2.54e-07
	LS-Chebyshev		9.95e-07	1.63e-07	9.17e-07	1.43e-07	9.75e-07	1.38e-07	1.27e-06	1.36e-07	1.75e-06	1.33e-07
Oscillatory $Q = 50$	Smolyak		9.12e+00	1.08e+00	5.19e+00	5.72e-01	3.57e+00	2.89e-01	9.50e-01	1.09e-01	2.64e-01	3.52e-02
	LS-Uniform		7.55e+00	9.40e-01	8.38e+00	6.01e-01	7.17e+00	2.77e-01	5.27e+00	8.04e-02	1.64e+00	2.04e-02
	LS-Chebyshev		4.54e+00	9.56e-01	7.02e+00	5.28e-01	2.40e+00	2.16e-01	1.39e+00	5.20e-02	2.80e-01	9.82e-03
Prod. Peak $Q = 50$	Smolyak		9.27e-03	1.70e-03	2.81e-03	3.66e-04	4.17e-04	6.84e-05	7.91e-05	1.14e-05	1.20e-05	1.56e-06
	LS-Uniform		1.23e-02	1.67e-03	6.63e-03	2.38e-04	1.13e-03	3.83e-05	2.50e-04	5.41e-06	4.56e-05	7.19e-07
	LS-Chebyshev		1.02e-02	1.75e-03	3.78e-03	2.53e-04	3.15e-04	3.31e-05	4.34e-05	3.70e-06	7.90e-06	3.57e-07
Ridge Prod. $Q = 50$	Smolyak		6.15e+00	8.05e-01	5.48e+00	5.01e-01	3.00e+00	2.60e-01	2.12e+00	1.30e-01	1.18e+00	6.29e-02
	LS-Uniform		3.61e+00	5.34e-01	3.58e+00	2.63e-01	1.95e+00	1.36e-01	1.90e+00	7.09e-02	1.43e+00	3.74e-02
	LS-Chebyshev		2.91e+00	5.15e-01	2.04e+00	2.30e-01	1.31e+00	1.10e-01	6.90e-01	4.91e-02	3.62e-01	2.13e-02

TABLE 5. The errors $e_{\max}^{\text{wc}}$ and $e_{\text{mean}}^{\text{wc}}$ for $d = 6$, each scale for the different algorithms. Lowest value in bold.

			Scale 3		Scale 4		Scale 5		Scale 6		Scale 7	
			$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$
Bim. Gauss. $Q = 50$	Smolyak		7.98e-01	2.61e-01	7.87e-01	1.15e-01	5.11e-01	6.24e-02	3.20e-01	3.09e-02	2.41e-01	1.51e-02
	LS-Uniform		7.49e-01	1.03e-01	7.56e-01	6.24e-02	1.35e+00	4.22e-02	1.93e+00	2.18e-02	1.14e+00	1.20e-02
	LS-Chebyshev		6.65e-01	1.20e-01	4.30e-01	6.68e-02	2.53e-01	3.35e-02	2.29e-01	1.61e-02	1.29e-01	7.10e-03
Continuous $Q = 50$	Smolyak		2.20e-02	5.52e-03	9.98e-03	1.91e-03	6.65e-03	6.62e-04	4.59e-03	2.38e-04	1.61e-03	1.03e-04
	LS-Uniform		4.10e-02	4.89e-03	2.24e-02	2.02e-03	1.77e-02	8.28e-04	9.62e-03	3.38e-04	6.29e-03	1.41e-04
	LS-Chebyshev		2.30e-02	5.66e-03	1.03e-02	2.06e-03	5.43e-03	7.52e-04	3.13e-03	2.83e-04	1.57e-03	9.98e-05
Corner Peak $Q = 50$	Smolyak		6.91e-03	7.19e-04	1.43e-01	6.21e-03	1.25e-01	7.51e-03	1.72e-01	6.07e-03	1.58e-01	3.51e-03
	LS-Uniform		1.25e-02	8.00e-04	2.53e-02	5.68e-04	4.08e-02	4.40e-04	7.54e-02	5.05e-04	5.62e-02	2.16e-04
	LS-Chebyshev		7.88e-03	1.42e-03	8.90e-03	7.98e-04	3.83e-02	5.16e-04	7.42e-02	4.99e-04	2.40e-02	1.38e-04
Discont. $Q = 50$	Smolyak		2.81e+02	2.30e+01	3.75e+02	1.86e+01	5.28e+02	1.48e+01	4.92e+02	1.11e+01	3.68e+02	9.72e+00
	LS-Uniform		2.64e+02	3.04e+01	3.00e+02	2.37e+01	4.78e+02	1.97e+01	4.15e+02	1.50e+01	5.71e+02	1.31e+01
	LS-Chebyshev		2.03e+02	2.89e+01	2.99e+02	2.22e+01	3.28e+02	1.65e+01	3.66e+02	1.19e+01	3.06e+02	8.64e+00
Gaussian $Q = 50$	Smolyak		6.84e-01	1.48e-01	7.29e-01	8.68e-02	5.52e-01	5.32e-02	3.25e-01	3.65e-02	1.91e-01	1.40e-02
	LS-Uniform		7.19e-01	1.07e-01	5.81e-01	7.80e-02	7.53e-01	4.91e-02	9.35e-01	2.61e-02	9.85e-01	1.13e-02
	LS-Chebyshev		5.33e-01	1.02e-01	4.29e-01	7.14e-02	3.12e-01	4.27e-02	1.94e-01	1.98e-02	9.88e-02	7.25e-03
Geo. Mean $Q = 50$	Smolyak		1.31e-01	2.53e-02	4.70e-02	7.43e-03	1.16e-02	1.26e-03	2.83e-03	1.23e-04	4.68e-04	1.16e-05
	LS-Uniform		1.77e-01	1.59e-02	4.31e-02	4.03e-03	1.87e-02	7.63e-04	2.64e-03	7.89e-05	3.90e-04	6.39e-06
	LS-Chebyshev		1.07e-01	1.91e-02	4.13e-02	4.73e-03	1.42e-02	8.81e-04	1.82e-03	7.15e-05	1.66e-04	4.52e-06
Noise $Q = 50$	Smolyak		4.55e-06	8.46e-07	6.68e-06	1.09e-06	1.07e-05	1.55e-06	1.85e-05	1.97e-06	2.61e-05	2.63e-06
	LS-Uniform		1.43e-06	1.48e-07	1.80e-06	1.51e-07	5.48e-06	1.69e-07	8.07e-06	1.95e-07	1.36e-05	2.32e-07
	LS-Chebyshev		8.02e-07	1.69e-07	1.01e-06	1.48e-07	1.59e-06	1.46e-07	2.73e-06	1.40e-07	2.46e-06	1.37e-07
Oscillatory $Q = 50$	Smolyak		1.06e+01	1.15e+00	8.53e+00	7.93e-01	6.70e+00	4.91e-01	2.96e+00	2.58e-01	1.28e+00	1.05e-01
	LS-Uniform		9.02e+00	1.09e+00	1.39e+01	7.83e-01	1.87e+01	5.44e-01	1.07e+01	2.26e-01	8.43e+00	7.49e-02
	LS-Chebyshev		6.32e+00	1.15e+00	6.93e+00	7.68e-01	6.43e+00	3.93e-01	3.89e+00	1.55e-01	2.33e+00	4.39e-02
Prod. Peak $Q = 50$	Smolyak		1.13e-02	1.72e-03	3.79e-03	2.89e-04	5.02e-04	5.02e-05	1.10e-04	7.81e-06	1.64e-05	1.19e-06
	LS-Uniform		2.18e-02	1.57e-03	6.98e-03	2.34e-04	1.44e-03	3.32e-05	3.64e-04	4.48e-06	9.59e-05	6.50e-07
	LS-Chebyshev		9.40e-03	1.92e-03	2.40e-03	2.34e-04	4.85e-04	2.96e-05	8.24e-05	3.17e-06	1.44e-05	3.45e-07
Ridge Prod. $Q = 50$	Smolyak		2.00e+01	1.34e+00	1.25e+01	8.94e-01	9.74e+00	5.36e-01	4.65e+00	2.83e-01	2.73e+00	1.40e-01
	LS-Uniform		6.74e+00	8.14e-01	5.74e+00	4.82e-01	4.54e+00	2.65e-01	3.78e+00	1.46e-01	2.82e+00	8.19e-02
	LS-Chebyshev		8.29e+00	1.11e+00	4.53e+00	4.97e-01	2.40e+00	2.33e-01	1.77e+00	1.10e-01	8.24e-01	5.04e-02

TABLE 6. The errors $e_{\max}^{\text{wc}}$ and $e_{\text{mean}}^{\text{wc}}$ for $d = 7$, each scale for the different algorithms. Lowest value in bold.

		Scale 3		Scale 4		Scale 5		Scale 6	
		$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$
Bim. Gauss. $Q = 50$	Smolyak	5.56e-01	1.04e-01	5.81e-01	8.40e-02	4.34e-01	4.80e-02	2.98e-01	2.98e-02
	LS-Uniform	6.78e-01	7.45e-02	9.50e-01	4.69e-02	1.58e+00	3.14e-02	1.71e+00	2.25e-02
	LS-Chebyshev	6.95e-01	7.63e-02	4.35e-01	4.84e-02	3.40e-01	2.79e-02	2.32e-01	1.58e-02
Continuous $Q = 50$	Smolyak	2.18e-02	5.14e-03	1.51e-02	1.99e-03	5.89e-03	7.91e-04	4.40e-03	2.31e-04
	LS-Uniform	2.39e-02	5.54e-03	1.40e-02	2.23e-03	1.38e-02	8.73e-04	9.11e-03	3.50e-04
	LS-Chebyshev	2.12e-02	6.16e-03	1.13e-02	2.39e-03	5.62e-03	8.43e-04	3.66e-03	3.04e-04
Corner Peak $Q = 50$	Smolyak	3.56e-03	1.33e-04	6.71e-03	3.22e-04	2.51e-02	5.22e-04	1.07e-01	1.03e-03
	LS-Uniform	3.73e-03	1.80e-04	4.57e-03	8.64e-05	9.36e-03	8.50e-05	1.47e-02	8.28e-05
	LS-Chebyshev	2.15e-03	2.26e-04	3.63e-03	2.51e-04	3.88e-03	1.59e-04	7.60e-03	8.82e-05
Discont. $Q = 50$	Smolyak	5.96e+02	4.92e+01	5.86e+02	3.27e+01	6.62e+02	2.56e+01	7.96e+02	1.97e+01
	LS-Uniform	8.59e+02	5.69e+01	7.14e+02	3.85e+01	7.61e+02	3.06e+01	1.01e+03	2.46e+01
	LS-Chebyshev	9.36e+02	6.28e+01	5.35e+02	4.02e+01	7.10e+02	2.81e+01	6.59e+02	2.09e+01
Gaussian $Q = 50$	Smolyak	9.46e-01	2.18e-01	8.64e-01	1.13e-01	4.33e-01	5.79e-02	3.17e-01	2.92e-02
	LS-Uniform	4.95e-01	5.69e-02	6.33e-01	3.65e-02	8.86e-01	2.51e-02	1.25e+00	1.72e-02
	LS-Chebyshev	6.43e-01	7.48e-02	3.90e-01	4.15e-02	2.82e-01	2.48e-02	1.98e-01	1.30e-02
Geo. Mean $Q = 50$	Smolyak	4.04e-01	6.55e-02	2.23e-01	2.44e-02	1.00e-01	8.12e-03	3.84e-02	2.05e-03
	LS-Uniform	2.15e-01	1.81e-02	2.09e-01	1.00e-02	1.22e-01	3.46e-03	2.80e-02	9.89e-04
	LS-Chebyshev	2.11e-01	3.01e-02	1.75e-01	1.21e-02	8.49e-02	4.02e-03	2.08e-02	9.90e-04
Noise $Q = 50$	Smolyak	5.90e-06	8.98e-07	1.23e-05	1.48e-06	1.91e-05	2.04e-06	2.54e-05	2.76e-06
	LS-Uniform	1.10e-06	1.39e-07	1.65e-06	1.40e-07	2.68e-06	1.61e-07	5.59e-06	1.80e-07
	LS-Chebyshev	7.93e-07	1.63e-07	1.13e-06	1.55e-07	1.23e-06	1.47e-07	1.45e-06	1.43e-07
Oscillatory $Q = 50$	Smolyak	1.63e+01	1.69e+00	1.72e+01	1.14e+00	1.34e+01	7.83e-01	7.78e+00	4.42e-01
	LS-Uniform	8.15e+00	1.10e+00	1.23e+01	9.36e-01	2.15e+01	7.03e-01	1.96e+01	3.87e-01
	LS-Chebyshev	6.22e+00	1.24e+00	7.11e+00	9.91e-01	1.08e+01	6.46e-01	6.35e+00	3.18e-01
Prod. Peak $Q = 50$	Smolyak	1.16e-02	1.70e-03	3.83e-03	3.68e-04	8.88e-04	7.52e-05	1.83e-04	1.32e-05
	LS-Uniform	1.94e-02	1.90e-03	5.74e-03	3.60e-04	2.59e-03	6.62e-05	4.75e-04	8.53e-06
	LS-Chebyshev	1.11e-02	2.31e-03	3.49e-03	4.12e-04	1.38e-03	5.81e-05	1.59e-04	6.78e-06
Ridge Prod. $Q = 50$	Smolyak	2.59e+01	1.52e+00	2.89e+01	1.15e+00	1.68e+01	7.26e-01	1.23e+01	4.47e-01
	LS-Uniform	8.12e+00	1.10e+00	8.12e+00	5.94e-01	4.78e+00	3.34e-01	5.71e+00	1.75e-01
	LS-Chebyshev	7.20e+00	1.28e+00	5.92e+00	6.63e-01	5.18e+00	3.20e-01	3.38e+00	1.46e-01

TABLE 7. The errors $e_{\max}^{\text{wc}}$ and $e_{\text{mean}}^{\text{wc}}$ for $d = 8$, each scale for the different algorithms. Lowest value in bold.

		Scale 3		Scale 4		Scale 5		Scale 6	
		$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$
Bim. Gauss. $Q = 50$	Smolyak	5.64e-01	1.24e-01	5.72e-01	8.98e-02	4.82e-01	5.27e-02	3.84e-01	3.08e-02
	LS-Uniform	4.84e-01	5.22e-02	1.23e+00	3.54e-02	1.14e+00	2.41e-02	1.83e+00	1.60e-02
	LS-Chebyshev	6.72e-01	6.55e-02	4.68e-01	3.50e-02	3.52e-01	2.26e-02	2.48e-01	1.25e-02
Continuous $Q = 50$	Smolyak	2.61e-02	4.32e-03	1.23e-02	1.83e-03	5.99e-03	6.36e-04	2.66e-03	2.31e-04
	LS-Uniform	2.27e-02	4.38e-03	1.71e-02	1.98e-03	1.03e-02	7.58e-04	6.79e-03	2.99e-04
	LS-Chebyshev	2.49e-02	5.49e-03	1.11e-02	2.16e-03	5.76e-03	7.67e-04	2.45e-03	2.69e-04
Corner Peak $Q = 50$	Smolyak	6.22e-04	1.81e-05	2.19e-03	4.40e-05	9.84e-03	1.89e-04	2.74e-02	3.78e-04
	LS-Uniform	5.96e-04	2.28e-05	1.75e-03	4.01e-05	6.45e-03	4.19e-05	5.86e-03	2.72e-05
	LS-Chebyshev	4.85e-04	8.03e-05	1.45e-03	8.92e-05	4.63e-03	4.70e-05	3.15e-03	3.15e-05
Discont. $Q = 50$	Smolyak	1.01e+03	8.59e+01	1.38e+03	7.32e+01	1.40e+03	5.76e+01	1.64e+03	4.19e+01
	LS-Uniform	1.19e+03	9.05e+01	1.24e+03	7.74e+01	1.28e+03	6.15e+01	1.64e+03	5.16e+01
	LS-Chebyshev	1.31e+03	1.17e+02	1.46e+03	9.21e+01	1.45e+03	6.45e+01	1.43e+03	4.60e+01
Gaussian $Q = 50$	Smolyak	8.99e-01	1.92e-01	9.52e-01	1.07e-01	6.14e-01	5.86e-02	4.03e-01	3.20e-02
	LS-Uniform	5.37e-01	4.78e-02	4.95e-01	3.38e-02	8.07e-01	1.90e-02	1.37e+00	1.15e-02
	LS-Chebyshev	6.74e-01	5.36e-02	5.34e-01	3.47e-02	3.73e-01	1.89e-02	2.03e-01	9.52e-03
Geo. Mean $Q = 50$	Smolyak	7.41e-02	1.59e-02	2.26e-02	3.04e-03	6.07e-03	4.82e-04	1.51e-03	7.50e-05
	LS-Uniform	1.21e-01	1.05e-02	3.67e-02	1.91e-03	6.13e-03	2.28e-04	1.64e-03	2.57e-05
	LS-Chebyshev	8.00e-02	1.29e-02	2.42e-02	2.23e-03	3.60e-03	2.41e-04	5.74e-04	2.48e-05
Noise $Q = 50$	Smolyak	6.94e-06	1.20e-06	1.41e-05	1.78e-06	2.37e-05	2.53e-06	4.70e-05	3.65e-06
	LS-Uniform	1.06e-06	1.28e-07	2.31e-06	1.36e-07	3.43e-06	1.52e-07	6.67e-06	1.75e-07
	LS-Chebyshev	9.14e-07	1.68e-07	1.11e-06	1.54e-07	2.03e-06	1.49e-07	2.03e-06	1.47e-07
Oscillatory $Q = 50$	Smolyak	2.13e+01	2.42e+00	2.41e+01	1.55e+00	3.52e+01	1.17e+00	2.21e+01	7.35e-01
	LS-Uniform	8.77e+00	1.15e+00	1.67e+01	1.08e+00	3.29e+01	8.95e-01	4.55e+01	6.32e-01
	LS-Chebyshev	6.05e+00	1.35e+00	1.19e+01	1.15e+00	9.81e+00	8.84e-01	1.54e+01	5.40e-01
Prod. Peak $Q = 50$	Smolyak	7.21e-03	1.21e-03	3.29e-03	2.39e-04	8.71e-04	5.04e-05	2.18e-04	1.12e-05
	LS-Uniform	6.95e-03	7.63e-04	4.37e-03	1.84e-04	1.82e-03	3.80e-05	1.28e-03	7.63e-06
	LS-Chebyshev	9.66e-03	1.06e-03	2.23e-03	2.28e-04	5.74e-04	3.79e-05	1.77e-04	5.82e-06
Ridge Prod. $Q = 50$	Smolyak	3.16e+01	2.35e+00	4.64e+01	1.81e+00	4.10e+01	1.24e+00	2.88e+01	8.29e-01
	LS-Uniform	1.13e+01	1.57e+00	2.34e+01	9.65e-01	1.44e+01	5.40e-01	9.33e+00	2.99e-01
	LS-Chebyshev	1.35e+01	2.55e+00	3.32e+01	1.21e+00	1.68e+01	5.54e-01	7.52e+00	2.67e-01

TABLE 8. The errors $e_{\max}^{\text{wc}}$ and $e_{\text{mean}}^{\text{wc}}$ for $d = 9$, each scale for the different algorithms. Lowest value in bold.

		Scale 3		Scale 4		Scale 5		Scale 6	
		$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$	$e_{\max}^{\text{wc}}$	$e_{\text{mean}}^{\text{wc}}$
Bim. Gauss. $Q = 50$	Smolyak	6.39e-01	1.27e-01	5.41e-01	8.83e-02	4.38e-01	7.04e-02	3.61e-01	4.50e-02
	LS-Uniform	5.78e-01	5.70e-02	7.48e-01	4.30e-02	1.11e+00	2.89e-02	1.86e+00	1.90e-02
	LS-Chebyshev	6.33e-01	5.62e-02	4.35e-01	4.11e-02	3.77e-01	2.83e-02	2.39e-01	1.56e-02
Continuous $Q = 50$	Smolyak	2.00e-02	3.74e-03	1.27e-02	1.42e-03	7.28e-03	5.52e-04	3.28e-03	2.17e-04
	LS-Uniform	2.47e-02	4.06e-03	2.01e-02	1.55e-03	1.01e-02	6.12e-04	6.02e-03	2.46e-04
	LS-Chebyshev	2.20e-02	4.64e-03	1.00e-02	1.78e-03	5.14e-03	6.27e-04	2.86e-03	2.25e-04
Corner Peak $Q = 50$	Smolyak	2.07e-04	5.64e-06	2.36e-04	2.90e-06	7.69e-04	1.48e-05	1.13e-02	9.47e-05
	LS-Uniform	2.02e-04	5.53e-06	2.74e-04	8.05e-06	5.55e-04	5.34e-06	3.20e-03	1.50e-05
	LS-Chebyshev	1.58e-04	2.38e-05	2.24e-04	2.31e-05	7.81e-04	1.59e-05	1.69e-03	1.17e-05
Discont. $Q = 50$	Smolyak	2.82e+03	1.78e+02	3.37e+03	1.38e+02	3.53e+03	1.09e+02	4.05e+03	8.62e+01
	LS-Uniform	2.51e+03	2.23e+02	2.76e+03	1.68e+02	4.43e+03	1.34e+02	4.16e+03	1.14e+02
	LS-Chebyshev	2.62e+03	3.08e+02	2.86e+03	2.03e+02	2.93e+03	1.52e+02	4.05e+03	1.12e+02
Gaussian $Q = 50$	Smolyak	5.23e-01	4.79e-02	4.92e-01	2.87e-02	4.09e-01	2.21e-02	3.20e-01	1.14e-02
	LS-Uniform	4.34e-01	3.33e-02	5.26e-01	2.29e-02	1.02e+00	1.48e-02	1.12e+00	8.78e-03
	LS-Chebyshev	5.07e-01	3.98e-02	4.63e-01	2.40e-02	3.33e-01	1.53e-02	2.18e-01	7.81e-03
Geo. Mean $Q = 50$	Smolyak	9.28e-02	1.66e-02	3.83e-02	4.21e-03	1.23e-02	1.07e-03	3.33e-03	2.45e-04
	LS-Uniform	8.69e-02	8.44e-03	2.01e-02	1.37e-03	1.47e-02	2.59e-04	4.33e-03	6.22e-05
	LS-Chebyshev	6.43e-02	1.09e-02	1.60e-02	1.88e-03	6.82e-03	2.94e-04	2.21e-03	6.53e-05
Noise $Q = 50$	Smolyak	1.22e-05	1.58e-06	2.09e-05	2.06e-06	3.46e-05	3.30e-06	6.18e-05	4.69e-06
	LS-Uniform	9.92e-07	1.29e-07	1.68e-06	1.32e-07	3.73e-06	1.47e-07	6.88e-06	1.66e-07
	LS-Chebyshev	9.57e-07	1.74e-07	1.11e-06	1.57e-07	1.64e-06	1.52e-07	1.86e-06	1.49e-07
Oscillatory $Q = 50$	Smolyak	4.37e+01	3.21e+00	6.56e+01	2.06e+00	4.67e+01	1.42e+00	4.98e+01	1.00e+00
	LS-Uniform	6.32e+00	1.13e+00	1.57e+01	1.11e+00	2.87e+01	1.04e+00	5.25e+01	8.35e-01
	LS-Chebyshev	6.10e+00	1.36e+00	8.10e+00	1.24e+00	1.22e+01	1.05e+00	1.52e+01	7.34e-01
Prod. Peak $Q = 50$	Smolyak	1.02e-02	1.66e-03	4.39e-03	3.63e-04	1.70e-03	7.55e-05	2.44e-04	1.50e-05
	LS-Uniform	1.54e-02	1.80e-03	6.05e-03	3.43e-04	3.03e-03	6.24e-05	8.10e-04	1.15e-05
	LS-Chebyshev	2.12e-02	2.24e-03	6.71e-03	4.28e-04	1.45e-03	6.70e-05	3.01e-04	1.04e-05
Ridge Prod. $Q = 50$	Smolyak	3.75e+01	2.48e+00	7.01e+01	2.06e+00	5.27e+01	1.54e+00	4.49e+01	1.03e+00
	LS-Uniform	1.41e+01	1.35e+00	1.79e+01	8.66e-01	1.28e+01	4.79e-01	9.06e+00	2.64e-01
	LS-Chebyshev	1.33e+01	2.13e+00	1.67e+01	1.07e+00	1.07e+01	5.36e-01	6.86e+00	2.61e-01

TABLE 9. The errors $e_{\max}^{\text{wc}}$ and $e_{\text{mean}}^{\text{wc}}$ for $d = 10$, each scale for the different algorithms. Lowest value in bold.